

EDUCATION

- **ABES Institute of Technology** Ghaziabad, India
Bachelor of Engineering in Computer Science and Engineering; Aggregate: 75.2% *Aug. 2018 - June 2022*

PROGRAMMING SKILLS

- **Languages:** Python, Golang, C++, C, Javascript
- **Framework:** Pytorch, Django, Flask, FastAPI, Gin, Celery, gRPC, Rabbitmq, Apache Airflow, Kafka
- **DevOps:** Amazon Web Services, Google Cloud, Docker, Kubernetes, ArgoCD, New Relic, Jenkins, Sentry
- **Version Control:** Git, Gitlab
- **Databases:** MySQL, PostgreSQL, MongoDB, Elasticsearch, Redis, Milvus, Aerospike
- **Testing:** Selenium, Scrappy, Playwright

WORK EXPERIENCE

- **TripJack** New Delhi, India
SDE-2 *April 2025 – Present*
 - Built a real-time hotel room mapping pipeline using **FAISS**, **Milvus Vector DB**, **LLMs (Ollama)**, and neural reranking to automatically map **100K+ room records** across suppliers, achieving **95% Top-1 accuracy**.
 - Developed a **Golang** microservice using **gRPC** to integrate **50+ hotel suppliers**, reducing supplier onboarding time from **4+ weeks to less than 1 week** and accelerating business integrations by **75%**.
 - Built a centralized **Orchestrator Service** in **Java Spring Boot** handling **millions of hotel search requests/day**, enforcing validation, routing, and failover logic to improve platform reliability and maintainability.
 - Optimized hotel search performance using **BitSet-based filtering**, caching, and efficient pagination, reducing response latency from **60s to under 200ms (300x improvement)** while supporting large-scale inventory searches.
 - Implemented distributed observability using **OpenTelemetry**, **Metrics**, **Tracing**, and **Log Correlation** across microservices, reducing production debugging time by **60%**.
 - Designed and deployed automated **Jenkins CI/CD** pipelines on **AWS EC2**, reducing deployment effort by **80%** and enabling reliable, repeatable releases.
 - Architected and developed **MCP (Model Context Protocol)** servers for **PostgreSQL**, **MongoDB**, and **Elasticsearch**, enabling AI agents to execute queries, analyze data, and troubleshoot systems across **millions of records** with natural-language interactions.
 - Built an AI-powered **PR Review MCP** integrating repository analysis, static code checks, and automated review comments, reducing manual review effort by **40%** and accelerating developer feedback cycles.
- **Trademo** Gurgoan, India
SDE-1 (Backend and Data) *June 2022 - April 2025*
 - Led the migration from a monolithic **Python**-based service to a **Golang microservices architecture**, significantly improving scalability, maintainability, and reducing latency by **50%**.
 - Architected and implemented a flexible **Access Control (ACL)** system enabling **role-based access control (RBAC)** with dynamic permission management for modular features.
 - Designed and deployed a **Retrieval-Augmented Generation (RAG)** system to classify **HS codes** using **Langchain** and a **vector database**, improving classification precision and speed.

- Built **30+ automated data pipelines** using **Python** and **Apache Airflow**, ensuring consistent data delivery and improving workflow efficiency.
- Integrated **RabbitMQ** and **Celery** in **Django** for bulk search classification with **multi-queue task groups**, increasing throughput by **50%** and reducing system load.
- Optimized **Elasticsearch** indices, mappings, and queries, leading to significant improvements in **search accuracy** and **response time**.
- Optimized high-traffic **Golang APIs**, enabling indexing of over **8 million pages** on Google, boosting company visibility and organic traffic.
- Integrated CI/CD pipelines using **GitLab** and **ArgoCD**, reducing deployment time by **30%** and achieving **zero-downtime releases**.

- **Artyvis Technologies Pvt. Ltd.**

Kolkata, India

Python Developer Intern

Dec 2021 - March 2022

- Successfully performed web scraping on more than 50 websites to gather relevant data using Python and popular scraping libraries such as BeautifulSoup and Scrapy.
- Developed and implemented over 10 data pipelines to automate the extraction, transformation, and loading (ETL) process, facilitating the seamless integration of data from various sources into MongoDB databases.

- **S759Labs Technology LLP**

Pune, India

Data Engineering Intern

March 2021 - June 2021

- Applied data extraction techniques to retrieve financial information from more than 5 company PDF balance sheets, extracting relevant data points such as revenue, expenses, and assets using Python and PDF parsing libraries.

- **TalentCruz**

Delhi, India

Backend Developer Intern

Jul. 2020 - Oct. 2020

- Started a venture when I was a sophomore.
- Integration of the front-end with Backend using Django and Developing ideas for features.
- Responsible for the development of the database management and other backend scripting tasks.
- Also, Integration of Payment APIs of razorpay.

PROJECTS

- **PostgreSQL Query Monitoring & Analysis Platform**

Skills: C, gRPC, Golang, PostgreSQL Internals

Built a PostgreSQL extension to monitor and collect query execution metrics in real time using PostgreSQL server hooks. The extension buffers query events and streams them to an external gRPC service for analysis, enabling slow query detection, workload monitoring, and database performance insights.

SELECTED COURSEWORK

- Deep Learning, Computer Vision, Machine Learning, Natural Language Processing (NLP), Software Engineering, Database Management System (DBMS)

EXTRA CURRICULAR ACTIVITIES

- Grand Finalist in **Smart India Hackathon 2020 (Software Edition)**
- Winner of **Internal Hackathon at ABESIT of SIH 2020 (Software Edition)**